Cyclotomic↔Coxeter mechanism v0.1 — the load-bearing gate: why chain length = h(B_2) = 4 (forcing toward, not yet forced)

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Cyclotomic↔Coxeter mechanism — the load-bearing gate

0. The gate (Cal #147)

Cal flagged: the chain length is independently 4 (q=2 cyclotomic number theory) and the Coxeter number is independently 4 (B_2 invariant). Their EQUALITY is striking but the connecting mechanism is ASSERTED, not derived. Until derived, chain-termination is MATCHED to Coxeter, not FORCED.

This v0.1 develops the forcing mechanism. Three convergent candidates; the rank=2 specialness is the key.

1. The two independent "4"s

1.1 Cyclotomic side (number theory)

Cal #139 chain {rank=2, N_c=3, n_C=5, g=7} = first 4 primes. Chain length 4 from: - Mersenne maximal prefix: M_2=3, M_3=7, M_5=31, M_7=127 all prime; M_11 = 2047 = 23·89 composite (first break) - q=2 Gaussian q-integers via K59 GF(2^g) = GF(128)

1.2 Geometric side (Coxeter)

B_2 root system: Coxeter number h(B_2) = 4. Multiple expressions: - h(B_2) = 4 (order of Coxeter element c = s_1 s_2) - |Φ+(B_2)| = rank·h/2 = 4 (number of positive roots) - Σ affine marks (1,2,1) = 4 - max exponent + 1 = 3 + 1 = 4

1.3 The equality to explain

chain length (cyclotomic, 4) = h(B_2) (geometric, 4). WHY?

2. The rank=2 specialness (key observation)

At rank = 2, multiple quantities COLLAPSE to 4:

h(B_2) = 2·rank = rank^2 = 4 (all equal at rank=2)

- h(B_n) = 2n generally; at n=rank=2: h = 4
- rank^2 = 4 at rank=2
- 2·rank = 4 at rank=2

These coincide ONLY at rank=2 (for B_n: 2n = n^2 ▯ n=2). This is the SAME rank=2 specialness as the Weinberg identity (rank+g=N_c^2 ▯ rank=2, per Weinberg Mechanism v0.1).

Substantive thread: rank = 2 is the load-bearing input making h = 2·rank = rank^2 = 4. The chain length, generation count, and Weinberg identity ALL collapse at rank=2.

3. Three convergent mechanism candidates

3.1 Candidate A — Positive roots

|Φ+(B_2)| = 4. Substrate commits along each positive root direction once per cycle → chain length = |Φ+| = 4.

The 4 positive roots: α_1, α_2, α_1+α_2, 2α_1+α_2. Chain levels {rank, N_c, n_C, g} ↔ 4 positive roots (one per root).

Forcing argument: if substrate's commitment cycle visits each positive root once, chain length = |Φ+(B_2)| = 4 FORCED. Gap: need to derive substrate commitment = positive-root traversal.

3.2 Candidate B — Affine marks

Affine B_2^(1) marks (a_0, a_1, a_2) = (1, 2, 1), sum = 4 = h.

The imaginary root δ = a_0 α_0 + a_1 α_1 + a_2 α_2 = α_0 + 2α_1 + α_2. The marks sum to h.

Forcing argument: substrate's commitment cycle = one full δ traversal = Σ marks = h = 4 steps. Gap: need to derive commitment cycle = δ traversal.

3.3 Candidate C — Coxeter element order

Coxeter element c = s_1 s_2 has order h(B_2) = 4. The orbit of any generic weight under c has length h = 4.

Forcing argument: substrate's commitment cycle = Coxeter element c application; one cycle = h = 4 steps (per ord(c) = 4). Gap: need to derive commitment cycle = Coxeter element.

3.4 The three candidates are unified by h(B_2) = 4

All three (positive roots, affine marks, Coxeter element order) = h(B_2) = 4. They are different views of the same Coxeter structure. The forcing mechanism is: substrate commitment cycle period = h(B_2) via one of these (likely all three are the same statement).

4. The cyclotomic↔geometric bridge (the actual gate)

4.1 What needs connecting

- Cyclotomic: chain length 4 from Mersenne maximal prefix (number theory)
- Geometric: h(B_2) = 4 (root system)

The bridge: WHY does the Mersenne-prefix break (M_11 composite) occur at exactly the Coxeter number?

4.2 Candidate bridge via GF(2^g)

Substrate operates on GF(2^g) = GF(128), g = 7. The multiplicative group is cyclic of order M_g = 127.

Key: g = 7 = h(B_2) + N_c = 4 + 3? = 7 ⍰. Or g = N_c^2 − rank = 7 (per Weinberg web).

The field GF(2^g) has g = 7 = (chain top element). The cyclotomic structure of GF(2^7): - The chain {2,3,5,7} are the first 4 primes ≤ g = 7 - There are exactly 4 primes ≤ 7: {2,3,5,7} - π(7) = 4 = h(B_2) (prime counting function at g)

Candidate bridge: chain length = π(g) = number of primes ≤ g = 4. And π(g) = π(7) = 4 = h(B_2).

So the bridge: chain = primes ≤ g; g = 7; π(7) = 4 = h(B_2). The cyclotomic chain length (primes ≤ g) equals the Coxeter number because g and h are related (g = N_c^2 − rank; h = 2·rank = rank^2; at rank=2, N_c=3: g=7, h=4, π(7)=4).

4.3 Honest assessment of the bridge

$\pi(7) = 4 = h(B\_2)$ is suggestive but: - $\pi(g) = \pi(7) = 4$ requires g = 7 (which is itself a BST primary) - Why $\pi(g) = h(B\_2)$? At g=7, h=4, $\pi(7)$=4. Is this forced or coincidental? - $\pi(n) = n/\ln(n)$ asymptotically; $\pi(7) = 4$ exactly. h(B_2) = 4. Their equality at g=7 may be coincidental.

This bridge (chain = $\pi(g)$ = h) is a CANDIDATE, not a derivation. The deeper forcing (why $\pi(g) = h(B\_2)$) is open.

5. Most promising forcing direction

5.1 Via rank=2 specialness

The cleanest path: rank = 2 forces h(B_2) = 2·rank = rank^2 = 4. If the chain length is independently tied to rank (e.g., chain length = rank^2 via Hua-Look 2^(rank^2) structure), then both = 4 via rank=2.

- h(B_2) = rank^2 = 4 (geometric, rank=2)
- Hua-Look normalization 2^(rank^2) = 16 (substrate; rank^2 = 4 appears)
- chain length = rank^2 = 4?

Forcing candidate: chain length = rank^2 (from Hua-Look 2^(rank^2) substrate structure) = h(B_2) (from B_2 root system) — BOTH equal 4 via rank=2. The bridge is rank=2.

5.2 This unifies with the over-determination cluster

Per Cal #147: Coxeter JOINS the over-determination cluster (h(B_2)=4, rank^2=4, Mersenne-prefix, Hua-Look, K59). The rank=2 specialness EXPLAINS the cluster: at rank=2, h(B_2) = rank^2 = 2·rank = 4, so the Coxeter (h=4), Hua-Look (rank^2=4), and chain (4) all coincide BECAUSE rank=2.

The over-determination is not a coincidence-pile — it's the rank=2 specialness manifesting through multiple structures. This is the unifying insight.

6. Status: matched → forcing-toward

6.1 What's now clearer

- The "4" coincidences (Coxeter, rank^2, 2·rank, chain, |Φ+|, affine marks, π(g)) ALL collapse at rank=2
- rank=2 is the load-bearing input
- The over-determination cluster is unified by rank=2 specialness (not independent coincidences)

6.2 What still gates FORCED

- Derive chain length = rank^2 (or = h) from substrate commitment cycle structure (multi-week)
- Derive WHY substrate has rank=2 (B_2) uniquely — this is Strong-Uniqueness Theorem territory (C1 rank=2 RATIFIED via T2435)
- The cyclotomic side (Mersenne prefix) ↔ geometric side (Coxeter) full bridge

6.3 Honest tier

Chain-termination: still MATCHED-toward-FORCED. The rank=2 specialness is a SUBSTANTIVE advance (unifies the over-determination cluster), but the full forcing (commitment cycle period = h derivation) remains multi-week.

Per Cal #147: Type C (level-crossing cyclotomic↔geometric), FRAMEWORK tier. This v0.1 advances the understanding (rank=2 unifies the cluster) but doesn't close the forcing gate.

7. Honest scope

What's RIGOROUS: - h(B_2) = 4 = 2·rank = rank^2 at rank=2 (Lie theory + arithmetic) - |Φ+(B_2)| = 4, affine marks sum = 4, ord(Coxeter element) = 4 (standard) - h^∨(B_2) = N_c = 3 (dual Coxeter) - $\pi(7) = 4$ (prime counting) - rank=2 collapses multiple quantities to 4 (verified)

What this v0.1 establishes substantively: - rank=2 specialness as the load-bearing input (h = 2·rank = rank^2 = 4) - Three convergent mechanism candidates (positive roots / affine marks / Coxeter element) - Over-determination cluster UNIFIED by rank=2 specialness (Cal #147 framing deepened) - Candidate bridge: chain = π(g) = h (suggestive)

What's FRAMEWORK / NOT yet FORCED: - Substrate commitment cycle period = h(B_2) (the forcing derivation; multi-week) - chain length = rank^2 from Hua-Look structure (candidate) - cyclotomic Mersenne-prefix ↔ Coxeter full bridge - Why π(g) = h(B_2) at g=7 (may be coincidental within rank=2 web)

Cal #147 Type C: level-crossing (cyclotomic ↔ geometric). FRAMEWORK. Advanced via rank=2 unification but forcing gate open.

Cal #27 STANDING: at peak-elegance (rank=2 unifies everything). Discipline: this is a SUBSTANTIVE structural advance, NOT a claim of forcing closure. Honest — matched-toward-forced; rank=2 specialness is the thread; multi-week derivation remains.

— Lyra, Cyclotomic↔Coxeter mechanism v0.1 filed (Keeper #1, THE GATE). KEY: rank=2 specialness — h(B_2) = 2·rank = rank^2 = 4 collapse only at rank=2, unifying the over-determination cluster (not independent coincidences). Three convergent mechanism candidates (positive roots / affine marks / Coxeter element). Candidate bridge chain = π(g) = h. Advances matched→forcing-toward; full forcing derivation (commitment cycle period = h) multi-week. Same rank=2 specialness underlies Weinberg identity.